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PI Name:	Kucik, Dennis F. M.D., Ph.D.		
Project Title:	Effect of high energy particle irradiation on adhesiveness of vascular endothelium and its consequences for atherosclerosis		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
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PI Email:	kucik@uab.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	205-934-0062
Organization Name:	University of Alabama at Birmingham		
PI Address 1:	Pathology		
PI Address 2:	1025 18th St S, CBSE room 239		
PI Web Page:			
City:	Birmingham	**State:**	AL
Zip Code:	35205-4831	**Congressional District:**	7
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: changed to Cardiovascular Alterations Team (from Radiation Effects Team) effective 5/1/2008, per K. Major/NSBRI (6/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kabarowski, Janusz (University of Alabama at Birmingham)		
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Task Description:

Inflammation plays a major role in radiation damage to tissues. Chronic, low-level vascular inflammation is a risk factor for atherosclerosis. Consistent with this, exposure to radiation from a variety of sources is associated with increased risk of heart disease and stroke.
Part of the inflammatory response to radiation is a change in the adhesiveness of the endothelial cells that line the blood vessels, triggering inappropriate accumulation of white blood cells, leading to later, damaging effects of inflammation. Although some studies have been done on the effects of gamma irradiation on vascular endothelium, the response of endothelium to the high energy radiation likely to be encountered in prolonged space flight has not been determined. A systematic characterization will allow prediction of pathological consequences, including atherosclerosis, reducing the uncertainty in risk prediction and pointing the way to possible countermeasures.

Our hypothesis is that high energy heavy ions alter the adhesive properties of vascular endothelium, and resultant vascular inflammation accelerates atherosclerosis. We are continuing to test this hypothesis using both isolated cells and whole mice to determine the effects on vascular endothelium and the consequences for development of atherosclerosis. The original objectives were:

Aim #1. Determine the effects of iron ion and proton irradiation vs. gamma irradiation on expression of key adhesion molecules on cultured endothelial cells, both at baseline and following cytokine stimulation.

Aim #2. Determine the consequences (for adhesion) of iron ion- and proton-irradiation-induced adhesion molecule expression and compare these to gamma irradiation using functional adhesion assays.

Aim #3. Determine whether radiation accelerates atherosclerosis in vivo using the well characterized ApoE mouse model.

Aim #4: Determine whether Tepoxalin reduces radiation-induced atherosclerosis.

Key Findings:

1) X-rays, protons and iron ions increase surface expression of specific adhesion molecules involved in monocyte-endothelial cell adhesion, as measured by flow cytometry, but not as much as typical inflammatory stimuli, such as TNF-alpha. Since endothelial cell adhesiveness is strongly increased in response to all types of radiation tested (see #2), current thinking is that mechanisms other than increased adhesion molecule expression are responsible for the adhesion change.

Impact: This constitutes completion of Aim #1. Expression of key adhesion molecules is only weakly induced by radiation damage, in contrast to the effect of most soluble inflammatory stimuli. This has led us to investigate other mechanisms for radiation-induced cell adhesion increases (see #4).

2) Adhesiveness of irradiated endothelial cells following X-ray, iron ion (56Fe) or proton irradiation was measured using a flow-chamber-based adhesion assay designed to mimic conditions in the bloodstream. In all cases, radiation increased adhesion. These adhesion increases were as pronounced as those produced by typical inflammatory stimuli, even though radiation-induced increases in adhesion molecule expression were typically much smaller.

Impact: This confirms the hypothesis that radiation alters adhesive properties of vascular endothelium, and represents progress toward completion of Aim #2. This has significant implications for astronaut health on prolonged deep-space missions, since increased vessel wall adhesiveness is an early step in development of atherosclerosis.

3) To better understand the mechanism of the increased adhesion, we have begun to identify the adhesion molecules responsible. Antibody blocking of particular integrins on leukocytes can modulate the pro-adhesive effect of radiation of endothelial cells, even if the radiation had only minor effects on surface expression of endothelial ligands for these integrins. In addition, we are beginning to identify signaling pathways that may be involved in activation of these integrins.

Impact: This provides an opportunity to identify possible therapeutic targets that may be amenable to countermeasures.

4) Animal experiments confirmed that radiation-induced changes have consequences for atherosclerosis. ApoE -/- mice were irradiated using collimated beams focused on the aortic arches and carotids. Iron-ion irradiation (600 MeV at 2 and 5 Gy) was found to accelerate development of atherosclerotic plaques in portions of the aortas and carotids of irradiated mice at 13 weeks post-irradiation as compared to un-irradiated controls. This confirms the hypothesis that components of cosmic radiation can accelerate atherosclerosis, and completes the original objective of Aim #3 for iron. Results of proton experiments are being analyzed.

Impact: 56Fe accelerates atherosclerosis and may pose a risk for astronaut health.

Proposed Research for the coming year:

1) Si-ion experiments to begin to address the dependence of both cell adhesiveness and atherosclerosis on LET will begin in the Spring, 2010 run. Both cultured aortic endothelial cells and apoE -/- mice will be irradiated at 1 and 5 Gy. A better understanding of how atherosclerotic effects depend on LET will permit a better estimate of the risk of the complex mix of radiation types in cosmic radiation.

2) Analysis of atherosclerosis in mice already irradiated will continue.

3) Efforts to identify signaling pathways that may be responsible for radiation's effect on endothelial cell adhesiveness will continue. Initial screening and preliminary identification will be done with X-ray irradiated human aortic endothelial cells at UAB. Any pathways identified will then be tested for Fe, Si, and/or protons at NRSL.

4) Results so far for both animal and cell experiments will be submitted for publication (at least 2 major publications).

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Among the non-cancer risks of radiation exposure are inflammatory diseases in general and atherosclerosis in particular. Increased risk of atherosclerosis is a known side effect of radiation exposure (generally gamma rays) on Earth, and increased endothelial adhesiveness (an inflammatory process) is an important component of the mechanism. In this project, we determine whether iron ions and/or proton radiation pose vascular disease risks and begin to understand the mechanism and dose response. With therapeutic proton irradiation becoming more prevalent, the findings concerning protons in particular will have implications for cancer therapy. Since all results will be compared to similar experiments using gamma- or x-rays, a greater understanding of the risks posed by terrestrial radiation will also result. Additionally, we will determine the feasibility of using in vitro adhesion assays measuring acute changes in endothelial cell adhesivity to predict the risk of longer-term vascular damage. Finally, we will determine the feasibility of selected countermeasures. Thus, knowledge and insights gained in this project may also be used to prevent, counter or ameliorate biological damage from terrestrial radiation sources such as nuclear weapons or radiation bioterrorism as well as adverse effects from therapeutic radiation.
Task Progress:	1) A flow-chamber based adhesion assay designed to mimic conditions in the bloodstream was used to measure adhesiveness of irradiated endothelial cells, following either iron ion or proton irradiation. We found that radiation-induced adhesiveness increases were as pronounced as those produced by typical inflammatory stimuli. The significance of this finding is that it clearly demonstrates that both iron and proton irradiation have the potential to increase vessel wall adhesiveness, which is an early step in development of atherosclerosis. Therefore, the potential consequences for astronaut health on prolonged deep-space missions represent a risk that warrants further investigation. 2) Our data indicate that both protons and iron ions can increase surface expression of specific adhesion molecules involved in monocyte-endothelial cell adhesion, as measured by flow cytometry, but only slightly at the doses that stimulate adhesion. These changes are much smaller than what is observed for typical inflammatory stimuli, such as TNF-alpha. Since endothelial cell adhesiveness is strongly increased in response to all types of radiation tested, current thinking is that mechanisms other than increased adhesion molecule expression are responsible for the adhesion change. 3) In order to better understand the mechanism of the increased adhesion, we have begun experiments to identify the adhesion molecules and ligands responsible. We find that antibody blocking of particular integrins on leukocytes can modulate the pro-adhesive effect of radiation of endothelial cells, even if the radiation had only minor effects on endothelial-cell ligand surface expression. 4) One mechanism of activating integrin adhesion without changing the expression of either the integrin or its ligand is via cell-signaling mechanisms. This year, we began to identify signaling pathways involved in radiation-induced endothelial cell adhesiveness increases. 5) Animal experiments confirmed that radiation-induced changes have consequences for atherosclerosis. ApoE -/- mice were irradiated using collimated beams focused on the aortic arches and carotids. Mice were given 600 MeV 56Fe at 2 and 5 Gy in Spring 2007 and 2008 and Fall 2009; and 200 MeV protons at 0.5 and 2 Gy in Fall 2007 and 2008. Mice were then sacrificed at 13 and 40 weeks post-irradiation to determine whether iron ion and/or proton radiation accelerated development of atherosclerosis. We found that radiation significantly accelerates development of atherosclerosis in this mouse model.
Bibliography Type:	Description: (Last Updated: 04/12/2018)
Articles in Peer-reviewed Journals	Khaled S, Gupta KB, Wu X, Yu T, Chang P, Kucik DF. "Iron ion irradiation alters adhesiveness of vascular endothelium." Gravitational and Space Biology. 2008 Jun;21(2):25-6. , Jun-2008
Articles in Peer-reviewed Journals	Kucik DF. "Measurement of adhesion under flow conditions." Current Protocols in Cell Biology. 2009 Jun;Chapter 9:Unit 9.6. PubMed PMID: 19499509 , Jun-2009